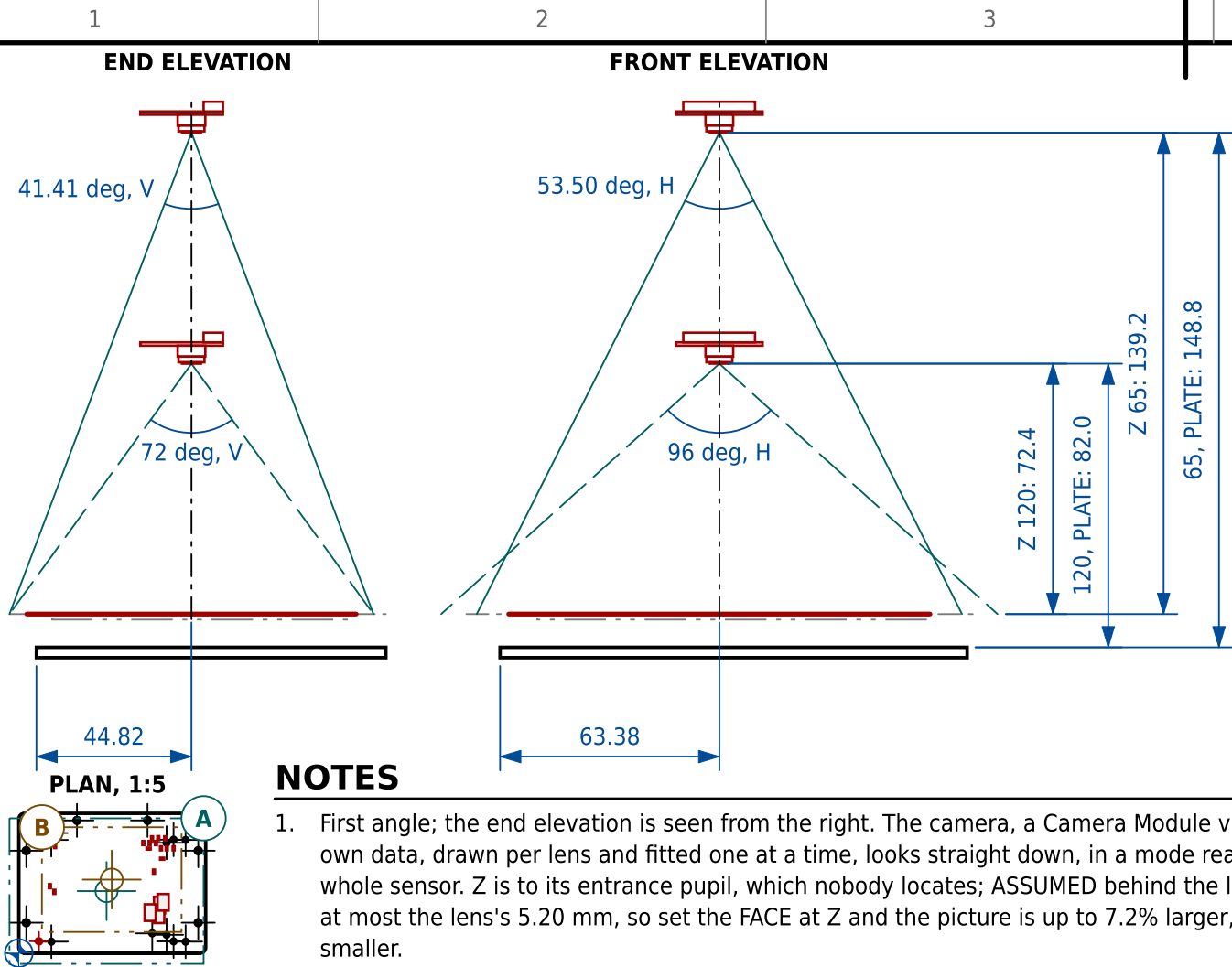




NOTES

1. First angle; the end elevation is seen from the right. The camera, a Camera Module v1.3 from its own data, drawn per lens and fitted one at a time, looks straight down, in a mode reading the whole sensor. Z is to its entrance pupil, which nobody locates; ASSUMED behind the lens face by at most the lens's 5.20 mm, so set the FACE at Z and the picture is up to 7.2% larger, never smaller.

NOTES (continued)

2. A frame is the smallest rectangle of the sensor's own 4:3 holding its target plus 5.00 mm all round, turned whichever way needs the lower camera; LONG says which. One margin absorbs both where the stand ends up -- 5.00 mm of lateral error, or a lean of 1.8 deg at frame A's 65 Z or 2.7 at its 120, measured at the picture's edge -- and what is not known about the lens, below: what one uses the other cannot.
3. PLANE, which Z is measured from: A/B from the DEMO BOARD's top face, not the plate's -- 8 mm standoffs and a 1.56 to 1.60 mm board put it 9.56 to 9.60 above the plate face; Z is from the higher. A stand set h mm too low covers only $(Z - h) / Z$ of what is drawn, so the error always loses the edges.
4. HEADROOM: a part standing on a board, anywhere in its envelope stays in frame A's picture up to 13.2 mm at 65, 6.9 mm at 120 above the plane Z is set from. Higher, raise the camera.
5. FOCUS. Both fixed lenses as sold are declared "Approx 1 m to infinity" (Raspberry Pi) and "Focus Distance 1 m to infinity" (Arducam), and every Z here is nearer: as sold, the subject is OUT OF FOCUS. At frame A's Z a point spreads to 21 px, 1.1 mm, at 65 and 19 px, 0.8 mm, at 120 (DERIVED, thin lens ASSUMED set at 2 m). Focused closer, Z is the higher of the close limit and $Z + f$ (DERIVED, thin lens); the autofocus module used here publishes NO close limit.

NOTES (continued)

6. The 120 deg lens is a fisheye, 120 deg on the diagonal: 96 x 72 is Arducam's split of that diagonal, equidistant, not a measurement. Its barrel distortion bows the picture's edges outwards on the board, so the frame's corners are inside it; the margin absorbs the narrower lenses the evidence allows, 120 diagonal, equisolid, 94.3 x 69.8, leaving 2.9 mm; Commonlands CIL282, scaled to 120, 94.4 x 70.8, leaving 3.9 mm; YXF4Y001A1, 1.79 mm, 92.4 x 73.9, leaving 6.9 mm. See RPICAM-LENS.
7. The "65 deg" name is the stock lens's DIAGONAL, which no vendor prints: $2 \times \arctan(\sqrt{3.6736^2 + 2.7384^2} / 2 / 3.60) = 64.94$ deg on OmniVision's image area, 64.42 on the pixel array. Every Z here is from the angle along each axis. 65 deg across frame A's long side would give Z 110.1 and lose 14.6 mm off each end.
8. Quotes are transliterated to ASCII: x, um and infinity for the signs the cached pages print.
9. Frame A, Every board, any revision: Every revision's assembled envelope -- outline, connectors, Pmod bodies -- in plate coordinates, in one: a camera fixed over the plate serves them all.
10. Frame B is the union of every LED and 7-segment over all five revision families. They are not clustered -- 90.91 x 62.79 of a 135 x 101 plate -- so frame B buys little over frame A.

FRAMES, AND Z TO FRAME THEM, mm

	FRAME	RECTANGLE	LONG	AXIS X	AXIS Y	Z 65	Z 120
A	Every board, any revision	140.27 x 105.20	X	63.38	44.82	139.2	72.4
B	Every LED and 7-seg	100.91 x 75.69	X	67.13	53.12	100.1	52.1

Z IN FOCUS, mm: THE LOWEST THAT FRAMES AND FOCUSES

	MODULE	CLOSE LIMIT	Z A	Z B
v1.3, lens unscrewed by hand	RPi Camera v1.3	60 mm, forum user	142.8	103.7
lens focused by hand	Arducam B0031	25.4 mm, DECL	142.8	103.7
motorised, the one used here	OV5647 AF-65	not published	--	--

LENSES, deg, AS USED: SEE RPICAM-LENS

	MODULE	H	V	DIAG	f mm	F	FOCUS FROM
65	RPi Camera v1.3	53.50	41.41	64.4 DERI	3.60 DECL	F2.9 DECL	Fixed, 1 m
AF	OV5647 AF-65	53.50	41.41	64.4 DERI	3.60 ASSU	F2.9 ASSU	Motor, not publ.
120	Arducam B006604	96	72	120.0 DECL	2.17 DERI	F2.4 ASSU	Fixed, 1 m

LEGEND

- Subject outline
- LED, display or connector
- Adjacent part or connector body
- Frame A's target, on the plane Z is measured from
- The 65 deg lens's field of view, 53.50 x 41.41, to its picture's edge
- The 120 deg lens's field of view, 96 x 72, to its picture's edge
- Frame A, the rectangle the picture must cover
- Frame B, the rectangle the picture must cover
- Dimension, extension and leader

SOURCES

1. Plate geometry: tinytapeout/mounting_plate/plate.py - Outline, placements and standoff; see TT-MP-PLATE.
2. Board and indicator positions: tinytapeout/boards.py - Outlines, connectors, Pmod bodies, LEDs and 7-segment displays, in plate coordinates.
3. Lens figures and focus: RPICAM-LENS - Every lens figure on this sheet, declared and derived, with the vendor pages it is quoted from.

CAMERA OVER THE TT MOUNTING PLATE

Camera Module OV5647, 65 and 120 degree lenses

DRAWN	VERSION	UNITS	SCALE
generated	v0.0-476-gf39d6ab	mm	1:2, PLAN 1:5
SIZE	SHEET	DRAWING NO	REV
A3	1 OF 1	RPICAM-OVER-PLATE	A
SUBJECT	GENERAL TOLERANCE	PROJECTION	
TT Mounting Plate	plate +/-0.20, LEDs +/-0.10, Z DERIVED		